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Discussion

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Given Electric field

step 1

$$\vec{E}(t) = \vec{E}_0(\omega)\cos\omega t \Rightarrow \boxed{\underbrace{\frac{\vec{E}_0(\omega)}{2}e^{-i\omega t}}_{P(t)} + \frac{\vec{E}_0^*(\omega)}{2}e^{i\omega t} = \vec{E}(t)}$$

In its simplest form:

step 2

$$\vec{P} = \varepsilon_0 \chi \vec{E} \Rightarrow \boxed{\vec{P}(\omega) = \varepsilon_0 \chi(\omega) E(\omega)}$$

$\frac{E_0(\omega)}{2}$

Polarization

Response

$$P(t) = \varepsilon_0 \chi(\omega)\frac{E_0(\omega)}{2}e^{-i\omega t} + \varepsilon_0 \chi(-\omega)\frac{\vec{E}_0^*(\omega)}{2}e^{i\omega t}$$

From Lecture class, we have

$$\chi^*(\omega) = \chi(-\omega)$$

Joule heat : $Q = \vec{j}\cdot\vec{E}$

current density

electric field

two approaches,

① $\vec{j} = \frac{\partial \vec{P}}{\partial t}$ $\qquad \vec{j} = \rho\vec{v}$

② $\rho = -\nabla\cdot\vec{P}$, $\qquad \nabla\cdot\vec{j} = -\frac{\partial\rho}{\partial t} = +\nabla\cdot\frac{\partial P}{\partial t}$

step 3 $\quad \vec{j} = \frac{\partial \vec{P}}{\partial t}$ $\qquad$ using $\vec{P}$ from above we get

$$\boxed{\vec{j} = -i\omega\varepsilon_0\chi(\omega)\frac{\vec{E}_0(\omega)}{2}e^{-i\omega t} + i\omega\varepsilon_0\chi^*(\omega)\frac{\vec{E}_0^*(\omega)}{2}e^{i\omega t}} \quad \text{— (A)}$$

Joule heat $Q = \vec{j} \cdot \vec{E}$

$$Q = \left(-i\omega\varepsilon_0 \chi(\omega) \frac{E_0(\omega)}{2} e^{-i\omega t} + i\omega\varepsilon_0 \chi^*(\omega) \frac{E_0^*(\omega)}{2} e^{i\omega t}\right) \cdot \left(\frac{E_0(\omega)}{2} e^{-i\omega t} + \frac{E_0^*(\omega)}{2} e^{i\omega t}\right)$$

$$= -i\omega\varepsilon_0 \chi(\omega) \frac{E_0^2(\omega)}{2} e^{-2i\omega t} - i\omega\varepsilon_0 \chi(\omega) \frac{E_0(\omega) E^*(\omega)}{4} + i\omega\varepsilon_0 \chi^*(\omega) \frac{E_0^*(\omega) \cdot E_0(\omega)}{2} + i\omega\varepsilon_0 \chi^*(\omega) \frac{E_0^{*2}(\omega) e^{2i\omega t}}{4}$$

we get 4 terms from this dot product. Take time average we get

step 4

dissipated heat: $\bar{Q} = -i\omega\varepsilon_0 \chi(\omega) \frac{1}{4} \underbrace{E_0(\omega) \cdot E_0^*(\omega)}_{|E_0(\omega)|^2} + i\omega\varepsilon_0 \chi^*(\omega) \frac{1}{4} |\vec{E}_0(\omega)|^2$

$$\bar{Q} = \frac{\omega\varepsilon_0}{4} |\vec{E}_0(\omega)|^2 \left(\underbrace{-i\underbrace{\chi(\omega)}_{\chi' + i\chi''} + i\underbrace{\chi^*(\omega)}_{\chi' - i\chi''}}\right)$$

$$-i(\chi' + i\chi'') + i(\chi' - i\chi'')$$

$$\rightarrow \cancel{-i\chi'} + \chi'' + \cancel{i\chi'} + \chi''$$

$2\chi''(\omega)$ so imaginary part survives.

$$\boxed{\bar{Q} = \frac{\omega\varepsilon_0}{2} |\vec{E}_0(\omega)|^2 \chi''(\omega)}$$

could have said,

$j(\omega) = \sigma(\omega) \vec{E}(\omega)$ where, $\sigma(\omega) = -i\omega\varepsilon_0 \chi(\omega)$

$E(\omega) = \frac{E_0(\omega)}{2} e^{-i\omega t}$

$$\mathrm{Re}[\sigma(\omega)] = -i\omega\varepsilon_0 \underbrace{\chi(\omega)}_{i\chi}$$

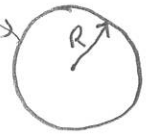
$$\Downarrow$$

$$\sigma'(\omega) = \omega\varepsilon_0 \chi''(\omega)$$

$\uparrow$ imaginary part survives

Now do a problem : metallic sphere

nanoparticle



$\vec{E}_{out}(\omega)\cos\omega t$

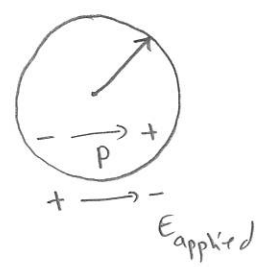
$$\varepsilon(\omega) = \varepsilon_0\left(1 - \frac{\omega_0^2}{\omega(\omega + i/\tau)}\right) \Rightarrow \chi(\omega) = -\frac{\omega_0^2}{\omega(\omega + i/\tau)}$$

$$\therefore \varepsilon(\omega) = \varepsilon_0(1 + \chi_e)$$

$$\boxed{E_{out}(t) = \frac{E_{out}(\omega)e^{-i\omega t}}{2} + \frac{E^*_{out}(\omega)e^{i\omega t}}{2}}$$

From previous Homework we know that, surrounding polarization creates the field, given by,

$$E_{ind} = -\frac{\vec{P}}{3\varepsilon_0}$$ This is induced electric field



$\vec{E}_{out}$ – applied field

$\vec{E}_{ind}$ – induced field

$\vec{E}_{in}$ – total field inside

we know,

$$\vec{P} = \varepsilon_0 \chi \vec{E}_{total}$$

so $$\vec{E}_{ind} = -\frac{\vec{P}}{3\varepsilon_0} = -\frac{1}{3\varepsilon_0}(\varepsilon_0 \chi E_{in}) = -\frac{1}{3}\chi E_{in}$$

$$\vec{E}_{in} = \vec{E}_{out} + \vec{E}_{ind}$$

$$\vec{E}_{in} = E_{out} - \frac{1}{3}\chi \vec{E}_{in}$$

$$E_{in}\left(1 + \frac{1}{3}\chi\right) = E_{out}$$

$$\boxed{E_{in} = \frac{1}{\left(1 + \frac{\chi}{3}\right)} \vec{E}_{out}}$$

$$\therefore \vec{P} = \varepsilon_0 \chi E_{in}$$

$$\boxed{\vec{P} = \varepsilon_0 \chi \left(\frac{1}{1 + \frac{\chi}{3}}\right) \vec{E}_{out} = \left(\frac{\varepsilon_0 \chi}{1 + \frac{\chi}{3}}\right) \vec{E}_{out}}$$

This is the static case

Now do frequency case

$$P(\omega) = \varepsilon_0 \chi(\omega) E(\omega)$$

$$\underbrace{\vec{P}(\omega)\, e^{-i\omega t}}_{\text{Fourier transform}} = \frac{\varepsilon_0 \chi(\omega)}{\left(1 + \frac{\chi(\omega)}{3}\right)} \vec{E}(\omega)\, e^{-i\omega t}$$

$$P(t) = \left(\frac{\varepsilon_0 \chi(\omega)}{1 + \frac{\chi(\omega)}{3}}\right) E(t) + \left(\frac{\varepsilon_0 \chi^*(\omega)}{1 + \frac{\chi^*(\omega)}{3}}\right) E^*(t)$$

plug in Electric fields

$$\boxed{P(t) = \left(\frac{\varepsilon_0 \chi(\omega)}{1 + \frac{\chi(\omega)}{3}}\right) \frac{E_{out}(\omega) e^{-i\omega t}}{2} + \left(\frac{\varepsilon_0 \chi^*(\omega)}{1 + \frac{\chi^*(\omega)}{3}}\right) \frac{E^*_{out}(\omega) e^{i\omega t}}{2}}$$

Ultimately we need dissipated energy so we need to do step 3 and 4

$$\vec{J} = \frac{\partial \vec{P}}{\partial t} \qquad Q = \vec{J} \cdot \vec{E}$$

$$\vec{J} = \frac{d\vec{P}}{dt} = \left(\frac{-i\omega\varepsilon_0 \chi(\omega)}{1+\frac{1}{3}\chi(\omega)}\right)\frac{E_{out}(\omega)}{2} e^{-i\omega t} + \left(\frac{i\omega\varepsilon_0 \chi^*(\omega)}{1+\frac{\chi(\omega)}{3}}\right)\frac{E_{out}(\omega)}{2} e^{i\omega t} \quad \text{—— (B)}$$

$\frac{dP}{dt} \Rightarrow$ same expression holds for (B) as (A) only different is

$$E_0(\omega) \Rightarrow \frac{E_{out}(\omega)}{1+\frac{\chi(\omega)}{3}}$$

in eq^n (A) → in eq^n (B)

$$\therefore \boxed{\bar{Q} = \frac{\omega\varepsilon_0}{2}\left|\frac{E_{out}(\omega)}{1+\frac{\chi(\omega)}{3}}\right|^2 \chi''(\omega)}$$

Re-write this with $\chi(\omega) = \chi'(\omega) + i\chi''(\omega)$

$$\bar{Q} = \frac{\omega\varepsilon_0}{2}\frac{|\vec{E}_{out}(\omega)|^2}{\left(1+\frac{\chi'(\omega)}{3}\right)^2 + \frac{1}{9}\chi''^2(\omega)}\chi''(\omega)$$

$$\mathrm{Im}\left(\frac{1}{1+\frac{\chi(\omega)}{3}}\right) = -\frac{\frac{1}{3}\chi''}{\left(1+\frac{1}{3}\chi'\right)^2 + \frac{1}{9}\chi''}$$

so we can say,

$$\bar{Q} = -\frac{\omega\varepsilon_0}{2}|\vec{E}_{out}(\omega)|^2 \,\mathrm{Im}\left(\frac{3}{1+\frac{\chi(\omega)}{3}}\right)$$

$$\bar{Q} = -\frac{3\omega\varepsilon_0}{2}\left|E_{out}(\omega)\right|^2 \operatorname{Im}\underbrace{\left(\frac{1}{1-\frac{\omega_0^2}{3\omega(\omega+i/\tau)}}\right)}_{*}$$

$$\because \chi(\omega) = -\frac{\omega_0^2}{\omega(\omega+i/\tau)}$$

$$* \quad \frac{1}{1-\frac{\omega_0^2}{3\omega(\omega+i/\tau)}} = \frac{3\omega(\omega+i/\tau)-\omega_0^2+\omega_0^2}{3\omega(\omega+i/\tau)-\omega_0^2}$$

$$= \frac{3\omega(\omega+i/\tau)}{3\omega^2-\omega_0^2+\frac{3\omega i}{\tau}} \times \frac{3\omega^2-\omega_0^2-\frac{3\omega i}{\tau}}{3\omega^2-\omega_0^2-\frac{3\omega i}{\tau}}$$

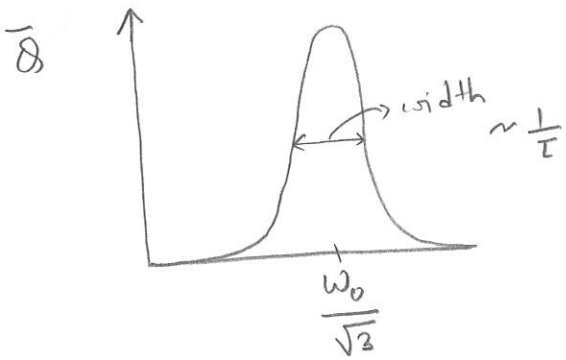
$$= \frac{9\omega^4-3\omega_0^2\omega^2-\cancel{\frac{9\omega^3 i}{\tau}}+\cancel{\frac{9\omega^3 i}{\tau}}-\frac{3\omega_0^2\omega i}{\tau}+\frac{9\omega^2}{\tau^2}}{(3\omega^2-\omega_0^2)^2+\left(\frac{3\omega}{\tau}\right)^2}$$

$$\operatorname{Im}\left(\frac{1}{1-\frac{\omega_0^2}{3\omega(\omega+i/\tau)}}\right) = \frac{-3\omega_0^2\omega}{\tau(3\omega^2-\omega_0^2)^2+\left(\frac{3\omega}{\tau}\right)^2}$$

plug back into $\bar{Q}$

$$\bar{Q} = \frac{3\omega\varepsilon_0}{2}\,\frac{3\omega_0^2\omega}{\tau(3\omega^2-\omega_0^2)^2+\left(\frac{3\omega}{\tau}\right)^2}\left|\vec{E}_{out}(\omega)\right|^2$$

$$\bar{Q} \approx \frac{9\varepsilon_0\omega^2\omega_0^2}{2\tau(3\omega^2-\omega_0^2)^2}\left|E_{out}(\omega)\right|^2 \quad \text{as } \tau \to \infty$$



<u>from a while back</u>

$$\varepsilon_0 (l+1) + l\,\varepsilon(\omega) = 0 \qquad l = 1, 2, 3 \ldots$$

$l = 1 \qquad \omega = \frac{\omega_0}{\sqrt{3}}$

$l = \infty \qquad \omega = \frac{\omega_0}{\sqrt{2}}$

* Look back at E_{in}

$$\vec{E}_{in} = \frac{1}{1+\frac{\chi}{3}} \vec{E}_{out}$$

geometric series,

$$\frac{1}{1+\chi/3} = 1 - \frac{\chi}{3} + \left(\frac{\chi}{3}\right)^2 - \left(\frac{\chi}{3}\right)^3 + \ldots$$

$$E_{in} = \vec{E}_{out} - \frac{\chi}{3} E_{out} + \frac{\chi}{3} E_{out} \frac{\chi}{3} + \ldots$$

$$= \vec{E}_{out} - \frac{\varepsilon_0 \chi E_{out}}{3\varepsilon_0} + \frac{\varepsilon_0 \chi}{3\varepsilon_0} E_{out} \frac{\varepsilon_0 \chi}{3\varepsilon_0} + \ldots$$

have an ∞ series, but instead have form $\frac{1}{1+\chi/3}$

so can view it as a geometric series.